E-COMMERCE CUSTOMER CHURN ANALYSIS AND POWER BI DASHBOARD

A Report submitted

in partial fulfillment for the Degree of

B. Tech.

in

Computer Science and Business Systems

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ABSTRACT

Customer churn, the loss of customers who stop buying from an online store, is costly because winning a new customer usually takes more effort and money than keeping an existing one. Most small online retailers, however, see churn only in reports prepared after the revenue has already been lost. This project builds a complete analytical pipeline that predicts which customers are likely to leave and presents the result in an interactive Power BI dashboard.

A synthetic data set of ten thousand customers and about fifty thousand orders covering eighteen months was generated and processed in Python. To avoid leaking the answer into the model, features were computed only from orders placed before a cutoff date, and each customer's churn label was determined from purchases made after it, using a waiting period matched to the customer's main product category. Logistic Regression, Random Forest and XGBoost classifiers were trained on the 5,181 customers with earlier purchase history and compared using cross-validation and a held-out test set. Random Forest ranked customers by risk most reliably, with an area under the receiver operating characteristic curve of 0.769 on the test set. Lowering its decision threshold to 0.40 matched the recall of Logistic Regression while giving higher precision.

The scored data were organised as a star schema and loaded into Power BI, where measures show that 35.26 percent of scored customers are predicted to churn, 450 are at high risk, and revenue worth 395.89 thousand dollars is at risk. Churn rises steadily from the most loyal to the lapsed customer segments, and engagement signals such as session count and cart abandonment are the strongest drivers. The report also states its limitations: the data are synthetic, and almost half of the customers have no earlier history from which to score them.

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CHAPTER 1
INTRODUCTION

1.1 Background and Motivation

Electronic commerce has grown from a supplementary retail channel into the main way in which many consumers buy goods. This growth has brought intense competition: a customer is only a click away from a rival that offers a lower price, faster delivery or a more personalised experience. In such a market the cost of winning a new customer is high. It is widely reported that retaining an existing customer is considerably cheaper, and that even small improvements in retention raise profit disproportionately [1]. Retailers have therefore turned their attention from acquisition alone to understanding, predicting and preventing customer attrition, commonly called churn.

In an online store churn rarely announces itself. Customers seldom close their accounts; instead their purchases become less frequent, their visits shorter and their carts are abandoned more often until they stop buying altogether. Rule-based reports detect this only after a long period of inactivity, by which time the opportunity for a low-cost win-back offer has usually passed.

Machine Learning (ML) and Business Intelligence (BI) address complementary parts of this problem. ML models the relationship between customer behaviour and the probability of future disengagement, producing a risk score for every customer before the loss appears in revenue. BI, here Microsoft Power BI, turns those scores into interactive dashboards that marketing, customer-support and management staff can use without data-science training. This project builds a complete pipeline that joins the two: from raw transactional and behavioural data, through a churn model in Python, to a dashboard on a star-schema data model.

1.2 Problem Statement

Most small and medium e-commerce organisations still rely on descriptive, backward-looking reporting to understand attrition, so customers disengage faster than the business notices. The specific difficulties are the following.

- **Lagging indicators:** monthly or quarterly churn reports describe what has already happened; by the time a spike is visible the affected customers have been inactive for a long time.
- **Undifferentiated retention spend:** without a risk score, discounts and offers go to the whole customer base, wasting budget on customers who would have stayed and missing those at genuine risk.
- **No value weighting:** the loss of an occasional low-value buyer is treated like the loss of a high-value repeat customer.
- **Fragmented data:** orders, browsing behaviour and support tickets are held in separate systems, so a single view of the customer is unavailable.

Existing dashboards add further gaps. They are descriptive rather than predictive; they apply one fixed churn definition (for example, no purchase in 90 days) although repurchase cycles differ widely between product categories; they are often built on flat extracts instead of a dimensional model; and they rarely quantify the revenue exposed to likely churners. This project addresses these gaps.

1.3 Objectives

The objectives of the project, and the outcome achieved for each, are given in Table 1.1.

Table 1.1: Objectives and outcomes

No.	Objective	Outcome achieved	Section
1	Predict the probability of churn of each customer from transactional and behavioural history.	Random Forest scores 5,181 customers; test-set area under the receiver operating characteristic curve (AUC-ROC) of 0.769.	6.1

No.	Objective	Outcome achieved	Section
2	Identify the behavioural drivers of churn.	Session count, cart abandonment and session duration rank highest.	5.6
3	Design a star-schema data model for consistent BI querying.	One fact table and four dimension tables.	4.6
4	Build an interactive Power BI dashboard.	Three pages: Executive Summary, Customer Segmentation and Model Insights.	6.3
5	Benchmark three classifiers and select one rigorously.	Cross-validation, test metrics and a threshold sweep.	6.1, 6.2
6	Quantify the financial impact of churn.	Revenue at Risk of $395.89K and a churn-adjusted lifetime value.	6.3

1.4 Scope of the Project

1.4.1 Target Stakeholders

Because the final deliverable is a decision-support dashboard rather than a stand-alone model, the scope follows the information needs of the stakeholders listed in Table 1.2.

Table 1.2: Stakeholders and their needs

Stakeholder	Primary need	Dashboard content used
Marketing team	Target high-risk, high-value segments with retention offers.	Churn by segment and channel; high-value customer list.
Customer support	Prioritise outreach to customers with early churn signals.	Customer-level churn probability and revenue.
Management	Monitor churn and revenue exposure at a glance.	Executive Summary indicators.
Data-science team	Validate and maintain the model.	Model Insights page.
Product team	Find categories where churn costs most.	Revenue at Risk by product category.

1.4.2 Inclusions and Exclusions

The project covers the generation and validation of a synthetic multi-table e-commerce dataset; data cleaning and feature engineering; snapshot-based churn

labelling with category-adjusted windows; training, evaluation and explanation of three classifiers; and the export of a star schema with a three-page Power BI dashboard.

It does not cover real-time data ingestion, deployment of the model as a production service, automated campaign execution, deep-learning sequence models, probabilistic lifetime-value forecasting or validation on real customer data. Some of these are discussed as future scope in Chapter 8.

1.5 Organisation of the Report

Chapter 2 reviews related work and identifies the gap addressed. Chapter 3 describes the dataset and the technologies used. Chapter 4 explains the methodology, from feature engineering to the star schema and the Data Analysis Expressions (DAX) measures. Chapter 5 summarises the exploratory analysis, and Chapter 6 presents the model results and the Power BI dashboard. Chapter 7 discusses applications and deployment on a live store, and Chapter 8 concludes the report with future scope.

CHAPTER 2
LITERATURE REVIEW

2.1 Customer Churn Prediction

Churn prediction has been studied extensively in the telecommunications industry, which pioneered many of the methods later applied to retail. Ahmad, Jafar and Aljoumaa [2] built a churn model for a telecom operator on a big-data platform, engineered features from customer records and compared several classifiers; XGBoost gave the best result, with an area under the curve of about 93 per cent. Vafeiadis et al. [3] compared neural networks, support vector machines, decision trees, Naive Bayes and logistic regression on a public telecom dataset and showed that boosting improved the performance of the individual classifiers. Amin et al. [4] studied churn classification in telecom data using a data-certainty approach.

These studies concentrate on predictive accuracy for telecom data. They tend to say little about how the predictions reach the business users who must act on them, how the churn label is defined without leaking the answer into the features, or how the financial exposure is expressed.

2.2 Techniques Used in This Project

Random Forest [5] builds many decision trees on bootstrap samples of the data and averages their votes; it copes well with non-linear relationships and outliers and provides a measure of feature importance. XGBoost [6] is an efficient implementation of gradient-boosted trees that adds trees one at a time to correct the errors of the previous ones. Logistic Regression, a linear model with interpretable coefficients, serves as the baseline.

Churners are normally a minority class, and classifiers trained on imbalanced data tend to favour the majority class [7]. The Synthetic Minority Over-sampling Technique (SMOTE) [8] creates new minority examples by interpolating between

existing ones; in this project it is applied to training data only. To explain predictions, SHAP [9] assigns each feature a contribution to every individual prediction based on Shapley values from game theory.

2.3 Business Intelligence and Dimensional Modelling

BI tools present data to decision-makers through interactive dashboards, and the speed and consistency of their figures depend on the data model beneath them. Dimensional modelling [10] separates measurable facts, such as order amounts, from the descriptive dimensions, such as customer, product and date, used to filter them. The result is the star schema adopted in this project.

2.4 Research Gap and Contribution

Academic studies optimise classification metrics in isolation, while industry BI templates optimise visual usability in isolation. Table 2.1 contrasts them with this project, which combines a leakage-safe churn model for an e-commerce setting with a governed dashboard that expresses the result as revenue at risk.

Table 2.1: Comparison with existing approaches

Aspect	Typical academic studies	Typical BI templates	This project
Domain	Mostly telecom	Generic samples	E-commerce
Prediction	Yes	No	Yes; three classifiers compared
Churn label	Often supplied with the data	Not applicable	Snapshot label with category-adjusted windows
Financial view	Rarely	Rarely	Revenue at Risk and churn-adjusted lifetime value
Business interface	Seldom addressed	Strong	Star schema and three-page dashboard

CHAPTER 3
DATASET AND TECHNOLOGIES USED

3.1 Dataset

3.1.1 Source and Generation

Public e-commerce datasets such as the UCI Online Retail II dataset [11] contain transactions but lack customer demographics, browsing behaviour, support interactions and signup dates, which the schema of this project requires. Combining a partial public dataset with simulated columns would be hard to defend, so a complete synthetic dataset was generated with NumPy and Faker using a fixed random seed (42), which makes it reproducible.

The data cover 18 months, January 2023 to June 2024, and contain 10,000 customers in six regions, 50,404 orders in eight product categories (200 products) and one behavioural summary record per customer. In all, 9,510 customers placed at least one order, and order dates always fall on or after the customer's signup date.

3.1.2 Raw Tables

Table 3.1: Raw tables generated

File	Rows	Key attributes
customers.csv	10,000	CustomerID, Age, Gender, Region (Central, East, North, Northeast, South, West), AcquisitionChannel (Organic, Paid Search, Social Media, Referral, Email), SignupDate
orders.csv	50,404	OrderID, CustomerID, OrderDate, ProductCategory, ProductID, ProductName, OrderAmount, Quantity, DiscountUsed, ReturnFlag
behaviour_support.csv	10,000	CustomerID, SessionCount_30d, CartAbandonmentRate, AvgSessionDuration, SupportTicketCount, LastSupportResolutionTime

3.1.3 Target Variable

The churn label is constructed rather than supplied. A snapshot (cutoff) design is used. All features are computed from orders strictly before the cutoff date of 1 January 2024, which is month 12 of the 18-month window. A customer is labelled churned (ChurnFlag = 1) if they place no order after the cutoff within a window set by their dominant product category, equal to 1.5 times the expected repurchase cycle of that category (Table 3.2).

Table 3.2: Category-adjusted churn windows

Category group	Expected cycle	Churn window	Window ends
Electronics Accessories, Groceries	60 days	90 days	1 Apr 2024
Clothing, Beauty, Books	90 days	135 days	16 May 2024
Home & Garden, Sports	105 days	157 days	7 Jun 2024
Furniture	120 days	180 days	29 Jun 2024

All windows end before the last date in the dataset (30 June 2024). The forward-looking label prevents leakage: if churn were defined from Recency and Recency were also a model input, the model would simply be given the answer.

3.1.4 Scored and Unscored Customers

Only customers with at least one order before the cutoff have features to compute. This gives 5,181 modelled (scored) customers, of whom 1,172 are labelled churned and 4,009 retained, a label churn rate of 22.6 per cent. The other 4,819 customers signed up after the cutoff or had no earlier orders; they stay in the dataset flagged IsScored = 0. This is the cold-start limitation common to behavioural churn models. The scored customers were split 80/20 with stratification; the test set holds 1,037 customers, 235 churned and 802 retained.

3.1.5 Limitations of the Dataset

Because the data are generated, the relationships a model can learn are those the generator built in. The accuracy figures describe performance on this dataset only and should not be read as expected performance on real retail data, where richer signals and different noise patterns would be present.

3.2 Technologies Used

The software and hardware used are listed in Tables 3.3 and 3.4. The main libraries are scikit-learn [12] for modelling, and Power BI with the DAX language [13], [14] for the data model and dashboard.

Table 3.3: Software used

Category	Tool	Purpose
Language	Python 3.12	Data generation, preprocessing, features, modelling
Data libraries	pandas, NumPy, Faker	Data manipulation and synthetic data
Machine learning	scikit-learn, XGBoost, imbalanced-learn, joblib	Training, evaluation, class balancing, model storage
Explainability	SHAP	Feature contributions
Visualisation	Matplotlib, Seaborn	Exploratory and evaluation plots
Business intelligence	Microsoft Power BI Desktop, DAX	Data model, measures, dashboard
Environment	VS Code, Google Antigravity IDE	Scripts run as cells, in order 01 to 05
Storage	CSV flat files	Raw, processed and star-schema tables

Table 3.4: Hardware used

Component	Minimum specification
Processor	Intel Core i5 or AMD Ryzen 5, or better
Memory	8 GB (16 GB recommended)
Storage	5 GB free, SSD preferred
Operating system	Windows 10/11, 64-bit (required by Power BI Desktop)

CHAPTER 4
PROJECT METHODOLOGY

4.1 Overview of the Pipeline

The project follows the five-stage pipeline in Table 4.1. It is implemented as five ordered scripts (01_eda to 05_star_schema_export) that call reusable modules for data generation, cleaning, features, labelling, modelling and star-schema export.

Table 4.1: Stages of the pipeline

Stage	Activities	Output
1. Data generation	Generate the raw tables; check completeness and referential integrity.	Three raw tables
2. Cleaning and features	Impute, de-duplicate, clip outliers; build features from pre-cutoff data.	customer_features.csv
3. Exploration and labelling	Distributions, correlations, cohorts; snapshot churn label.	EDA plots; ChurnFlag
4. Modelling	Train, cross-validate, tune threshold, explain.	Selected model; 5,181 scores
5. Star schema and Power BI	Export tables; define DAX measures; build dashboard.	Five CSV tables; .pbix file

4.2 Data Cleaning

Missing numeric values are imputed with the median and categorical values with the mode. Duplicate orders are removed, order amounts above the 99.5th percentile are capped, and date and numeric types are enforced. Referential integrity between the customer, order and behaviour tables is verified.

4.3 Feature Engineering

Features are computed only from orders before the cutoff date. In all, 24 features enter the models:

- **Recency-Frequency-Monetary (RFM):** Recency, Frequency, Monetary, the quintile scores R_Score, F_Score and M_Score, and the composite

RFM_Score. The RFM-based CustomerSegment is derived from these for the dashboard.

- **Purchasing behaviour:** AvgOrderValue, OrderAmountStd, AvgDaysBetweenOrders, UniqueCategories, DiscountUsageRate, ReturnRate and DominantCategory.
- **Engagement and support:** SessionCount_30d, CartAbandonmentRate, AvgSessionDuration, SupportTicketCount and LastSupportResolutionTime.
- **Customer profile:** Age, Gender, Region, AcquisitionChannel and DaysSinceSignup.

4.4 Churn Labelling and Leakage Control

The label is built as described in Section 3.1.3. As a check, the Random Forest feature-importance ranking was inspected. Recency ranks 13th of 24 features with an importance of 0.031, comparable to Region and AvgOrderValue. A leaked feature would dominate the ranking, so this confirms that the snapshot design separated Recency from the label.

4.5 Model Building and Evaluation

Churn prediction is a binary classification task. Three algorithms of increasing complexity were trained (Table 4.2) so that predictive power could be weighed against interpretability.

Table 4.2: Classifiers compared

Algorithm	Description	Role
Logistic Regression	Linear model with interpretable coefficients.	Baseline
Random Forest	Ensemble of decision trees on bootstrap samples.	Selected model
XGBoost	Gradient-boosted trees.	High-capacity comparison

The training partition is balanced with SMOTE, applied to training data only so that the test set keeps the natural class distribution. Models are compared by 5-fold stratified cross-validation (AUC-ROC) and on the held-out 20 per cent test set (AUC-ROC, precision, recall, F1 score, accuracy and confusion matrix). Recall on the churn class receives particular attention, because missing a genuine churner usually costs more than contacting a customer who would have stayed. All random processes (data generation, split, SMOTE, classifiers and cross-validation) use random_state = 42, so results are reproducible. Feature importance and SHAP values explain the selected model.

4.6 Star Schema and Power BI Model

The scored data are exported as a star schema (Table 4.3). Churn scores are customer-level facts, so they are held on Dim_Customer, one row per customer. Keeping them only on the order-level fact table would repeat each score on every order and inflate any sum or average.

Table 4.3: Tables of the star schema

Table	Grain	Key	Main columns
Dim_Customer	Customer (10,000)	CustomerID	Demographics, RFM_Score, CustomerSegment, ChurnProbability, ChurnFlag, PredictedCLTV, IsScored
Dim_Date	Day (546)	DateKey	FullDate, Year, Quarter, Month, MonthName, DayName
Dim_Product	Product (200)	ProductID	ProductName, ProductCategory
Dim_SupportTicket	Customer (10,000)	CustomerID	SupportTicketCount, LastSupportResolutionTime, CartAbandonmentRate
Fact_Orders	Order (50,404)	OrderID	CustomerID, DateKey, ProductID, OrderAmount, Quantity, ChurnFlag, IsScored

Four many-to-one, single-direction relationships link Fact_Orders to the dimensions: CustomerID to Dim_Customer and Dim_SupportTicket, DateKey to Dim_Date, and ProductID to Dim_Product. Dim_Date is a gap-free calendar from 1 January 2023 to 30 June 2024, marked as the date table. Because filters flow from the dimensions to the fact table only, the customer-level churn measures cannot be

sliced by order date, so the dashboard reports churn by customer attributes (segment, region, channel) and not by month.

The predicted lifetime value stored for each customer is a transparent heuristic, given in Equation 4.1.

$$PredictedCLTV = AvgOrderValue \times PredictedOrderFrequency(12\ months) \times (1 - ChurnProbability) \quad (4.1)$$

Customers that are not scored carry IsScored = 0 and the placeholder value −1; every churn measure filters on IsScored = 1 so that they cannot distort churn figures.

4.7 DAX Measures

The principal measures defined in the Power BI model are shown below. Total Revenue and Average Order Value are simple sum and average aggregations of Fact_Orders[OrderAmount].

```
Scored Customer Count =
  CALCULATE(DISTINCTCOUNT(Dim_Customer[CustomerID]),
            Dim_Customer[IsScored] = 1)

Churn Rate =
  DIVIDE(
    CALCULATE(DISTINCTCOUNT(Dim_Customer[CustomerID]),
              Dim_Customer[IsScored] = 1,
              Dim_Customer[ChurnFlag] = 1),
    CALCULATE(DISTINCTCOUNT(Dim_Customer[CustomerID]),
              Dim_Customer[IsScored] = 1))

Retention Rate = 1 - [Churn Rate]

High Risk Customers =
  CALCULATE(DISTINCTCOUNT(Dim_Customer[CustomerID]),
            Dim_Customer[IsScored] = 1,
            Dim_Customer[ChurnProbability] >= 0.70)

Revenue at Risk =
  CALCULATE(SUM(Fact_Orders[OrderAmount]),
            Fact_Orders[IsScored] = 1,
            Fact_Orders[ChurnFlag] = 1)

Average Churn Probability =
  CALCULATE(AVERAGE(Dim_Customer[ChurnProbability]),
            Dim_Customer[IsScored] = 1)

Avg Predicted CLTV =
  CALCULATE(AVERAGE(Dim_Customer[PredictedCLTV]),
```

```
        Dim_Customer[IsScored] = 1)
```

Revenue at Risk is the historical order value of scored customers whom the model flags as likely to churn. It measures current exposure, not a forecast of lost revenue.

4.8 Validation and Issues Resolved

Each stage was cross-checked against independent evidence, such as recomputed confusion matrices, row counts and saved metrics files. The checks found the defects in Table 4.4, all corrected before the final results were produced.

Table 4.4: Issues found and their resolution

Issue found	Effect	Resolution
Cutoff defaulted to month 6 instead of month 12.	Only 1,639 customers had pre-cutoff history.	Cutoff set to 1 Jan 2024; 5,181 customers modelled.
Unscored customers defaulted to ChurnFlag = 1.	Churn rate inflated from about 23% to about 44%.	IsScored flag and −1 placeholders; measures filter on IsScored = 1.
Evaluation table showed Random Forest figures in the Logistic Regression row.	Logistic Regression recall understated (51.5% against 68.5%).	Recomputed from confusion matrices and the saved metrics file.
Dim_Date built only from order dates.	Gaps prevented use as a date table.	Rebuilt as a contiguous 546-day calendar.
Churn scores stored on every order row.	Sums and averages inflated.	Scores moved to Dim_Customer.
ChurnFlag generated at the 0.5 threshold.	Inconsistent with the selected operating point.	Regenerated at threshold 0.40.

CHAPTER 5
EXPLORATORY DATA ANALYSIS SUMMARY

Exploratory analysis was carried out in the first notebook and covers univariate and bivariate distributions, a correlation heatmap, cohort retention, an order-frequency Pareto analysis, temporal trends and behavioural distributions; the plots are saved with the project. This chapter summarises the observations confirmed by the final dashboard and model outputs.

5.1 Data Quality and Coverage

All foreign keys in Fact_Orders resolve to their dimension tables and no primary or foreign key is null. The average order value is $70 and total revenue is $3.54 million over 50,404 orders, about five orders per customer over the 18 months.

5.2 Customer Base Composition

The RFM-based segments of the 5,181 scored customers are given in Table 5.1.

Table 5.1: Segments of scored customers

Segment	Customers	Share
Potential Loyalist	≈1,510	29.2%
Loyal	≈1,380	26.5%
Champion	≈930	18.0%
At Risk	≈850	16.4%
Lost	≈510	9.9%

5.3 Revenue Structure

Revenue is concentrated in a few categories. Furniture accounts for 35.4 per cent ($1.25 million), followed by Electronics Accessories (14.1 per cent, $0.50 million), Home & Garden (12.6 per cent, $0.45 million), Clothing (11.0 per cent, $0.39

million) and Sports (9.9 per cent, $0.35 million). Furniture also carries the largest revenue at risk.

5.4 Temporal Pattern

Daily revenue rises through the 18-month window and most steeply in the final months. This is consistent with customers signing up uniformly across the window, which enlarges the active base, and with the order-timing logic of the generator. It is a property of the synthetic data and should not be read as market growth. It is also the reason churn is analysed by customer attributes and not by calendar month.

5.5 Predicted Churn Patterns

Predicted churn varies little by region (about 33 to 37 per cent, Northeast highest and Central lowest) and by acquisition channel (average churn probability 0.31 to 0.34, Organic highest and Email lowest), which matches the low importance of these features. It varies strongly by RFM segment, rising from about 13 per cent for Champions to about 66 per cent for Lost customers.

5.6 Feature Importance

Table 5.2 gives the Random Forest feature importance (Gini) and the SHAP rank, where available, of the leading features and of Recency.

Table 5.2: Feature importance

Rank	Feature	Gini importance	SHAP rank (mean |SHAP|)
1	SessionCount_30d	0.152	1 (0.078)
2	CartAbandonmentRate	0.102	3 (0.042)
3	AvgSessionDuration	0.092	4 (0.037)
4	DaysSinceSignup	0.078	2 (0.053)
5	DominantCategory	0.042	—
6	RFM_Score	0.042	—
7	SupportTicketCount	0.042	—

Rank	Feature	Gini importance	SHAP rank (mean |SHAP|)
8	AvgDaysBetweenOrders	0.038	5 (0.022)
13	Recency	0.031	—

Both methods agree that engagement signals, meaning how often customers visit, how long they stay and how often they abandon carts, matter more than raw purchase timing. Recency ranks 13th, which supports the leakage check in Section 4.4.

CHAPTER 6
RESULTS AND DISCUSSION

6.1 Model Performance

Table 6.1 compares the three classifiers on the held-out test set (1,037 customers, 235 churned) at the default 0.5 threshold; the cross-validation AUC-ROC is the mean of five stratified folds. Table 6.2 gives the confusion matrices.

Table 6.1: Test-set and cross-validation results

Model	AUC-ROC test	AUC-ROC CV	Precision	Recall	F1	Accuracy
Logistic Regression	0.746	0.764 ± 0.018	0.386	0.685	0.494	0.682
Random Forest	0.769	0.777 ± 0.014	0.475	0.515	0.494	0.761
XGBoost	0.751	0.753 ± 0.013	0.452	0.540	0.492	0.747

Table 6.2: Confusion matrices at the 0.5 threshold

Model	True negatives	False positives	False negatives	True positives
Logistic Regression	546	256	74	161
Random Forest	668	134	114	121
XGBoost	648	154	108	127

At the default threshold the models behave differently. Logistic Regression flags about 40 per cent of customers and reaches the highest recall (68.5 per cent) at the lowest precision, while Random Forest is more conservative. Random Forest nevertheless has the best AUC-ROC on both the test set and cross-validation, which means it ranks customers by risk most reliably, independently of any threshold.

6.2 Threshold Tuning and Model Selection

To test whether the higher recall of Logistic Regression was an intrinsic advantage or only a product of its default threshold, the Random Forest threshold was swept

downward. At a threshold of 0.40 Random Forest matches the recall of Logistic Regression (about 69 per cent) while being more precise, as Table 6.3 shows.

Table 6.3: Comparison at equal recall

Configuration	Recall	Precision	F1	Customers flagged
Logistic Regression, threshold 0.5	≈69%	38.6%	0.494	417
Random Forest, threshold 0.40	≈69%	41.1%	0.516	397

Random Forest therefore dominates at the matched recall level and is selected as the final model, with the threshold set to 0.40 to favour recall, which suits a retention use case. The exported ChurnFlag and all dashboard churn measures use this threshold.

6.3 Power BI Dashboard

6.3.1 Page 1: Executive Summary

Table 6.4 lists the indicators on the first page, which is shown in Figure 6.1.

Table 6.4: Executive Summary indicators

Indicator	Value	Definition
Predicted churn rate	35.26%	Scored customers with probability ≥ 0.40, over scored customers
High-risk customers	450	Scored customers with probability ≥ 0.70
Revenue at Risk	$395.89K	Order value of scored customers flagged as churners
Avg. predicted CLTV	$376.60	Mean churn-adjusted lifetime value of scored customers
Average order value	$70	Mean order amount
Total revenue	$3.54M	All 50,404 orders

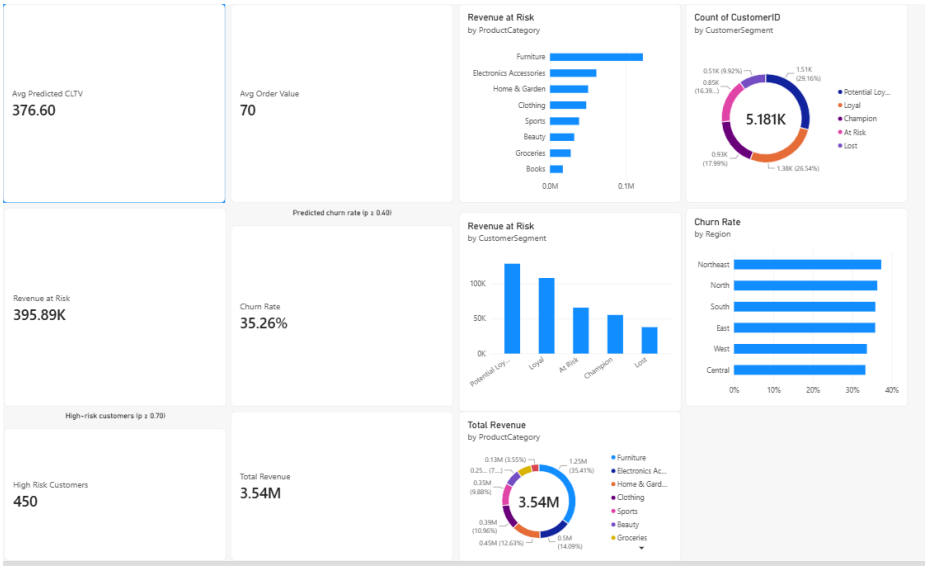


Figure 6.1: Power BI page 1: Executive Summary

The predicted churn rate of 35.26 per cent differs from the 22.6 per cent label rate of Section 3.1.4: the former is the share of customers the model flags at the 0.40 threshold, the latter is the historical share that actually stopped buying. The lower threshold deliberately flags more customers. Revenue at Risk is highest for Furniture, in line with its share of revenue. By segment it is largest for Potential Loyalists (about $130K) and Loyal customers (about $108K), followed by At Risk (about $65K), Champion (about $55K) and Lost customers (about $38K).

6.3.2 Page 2: Customer Segmentation

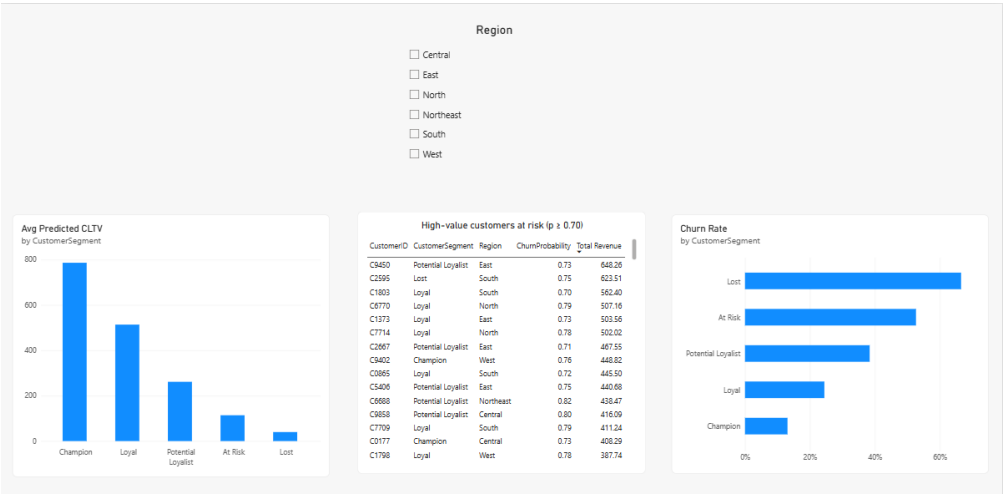


Figure 6.2: Power BI page 2: Customer Segmentation

Table 6.5: Approximate values read from Page 2

Segment	Predicted churn rate	Avg. predicted CLTV
Champion	≈ 13%	≈ $790
Loyal	≈ 24%	≈ $515
Potential Loyalist	≈ 38%	≈ $260
At Risk	≈ 52%	≈ $115
Lost	≈ 66%	≈ $40

Churn rises steadily from Champion to Lost, as expected. Average predicted CLTV falls in the same order because it is multiplied by (1 − churn probability); it is a churn-adjusted value, not raw spending. The customer list shows customers with churn probability of at least 0.70, sorted by total revenue, so a retention team can see which high-value customers are most worth contacting first.

6.3.3 Page 3: Model Insights

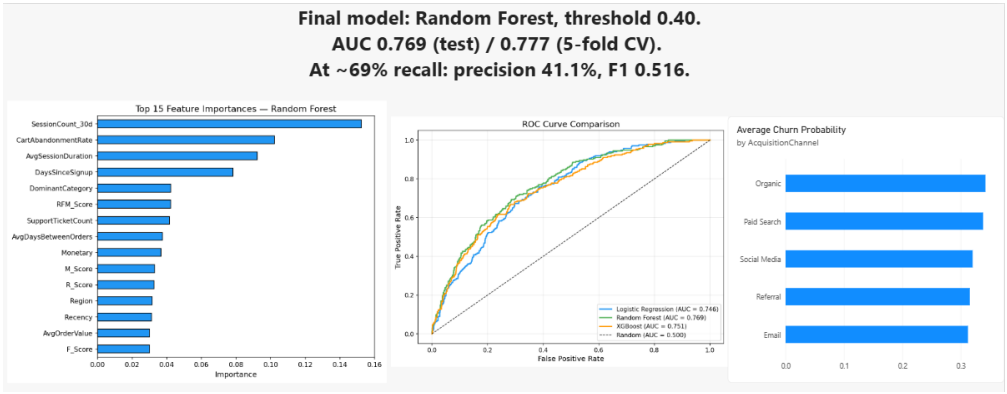


Figure 6.3: Power BI page 3: Model Insights

Page 3 combines the headline metrics of the selected model, the Random Forest feature-importance chart, the comparison of ROC curves (AUC 0.746, 0.769 and 0.751 for Logistic Regression, Random Forest and XGBoost) and a live chart of average churn probability by acquisition channel. The model figures are static images produced by the Python pipeline; the channel chart responds to the Power BI model.

6.4 Limitations

- **Synthetic data:** results describe this dataset only; real data would be noisier and richer, so the AUC-ROC would differ.
- **Cold start:** 4,819 customers (48 per cent) have no pre-cutoff history and cannot be scored; a separate model for new customers would be needed.
- **Moderate recall:** the selected configuration identifies roughly two thirds of true churners, and about 59 per cent of flagged customers are false alarms.
- **In-sample scoring:** the final model is applied to all scored customers, including those used for training, so their dashboard probabilities are optimistic. The reported metrics come only from the held-out test set.
- **Random validation split:** the test set is drawn from the same snapshot, so there is no out-of-time validation.
- **Heuristic measures:** both the lifetime value and Revenue at Risk are directional indicators, not financial forecasts.

CHAPTER 7
APPLICATIONS AND DEPLOYMENT

7.1 Applications

The system supports day-to-day decisions in an e-commerce organisation. The main uses are summarised in Table 7.1.

Table 7.1: Applications of the system

Application	Function	How it is used
Targeted retention campaigns	Marketing	Offer incentives to flagged high-value customers instead of the whole base.
Support prioritisation	Customer support	Contact customers with rising churn probability before they leave.
Revenue-exposure reporting	Management	Track Revenue at Risk as a standing indicator.
Category review	Product	Use revenue at risk by category to see where churn costs most.
Loyalty design	Marketing / CRM	Use RFM segments to design tiered benefits.
Channel evaluation	Marketing	Compare churn by acquisition channel when allocating budget.

The same approach, consisting of behavioural features, supervised churn classification, a star-schema model and DAX indicators, transfers to other subscription or transaction businesses such as software services, telecom, digital media and banking.

7.2 Deployment on a Live Store

The system built in this project is an offline, batch analytics prototype, developed and validated on a synthetic dataset. It is an internal decision-support tool for business teams, not for shoppers, so it would not sit inside the shopping site. Figure 7.1 shows how the pipeline would be deployed alongside a real store.

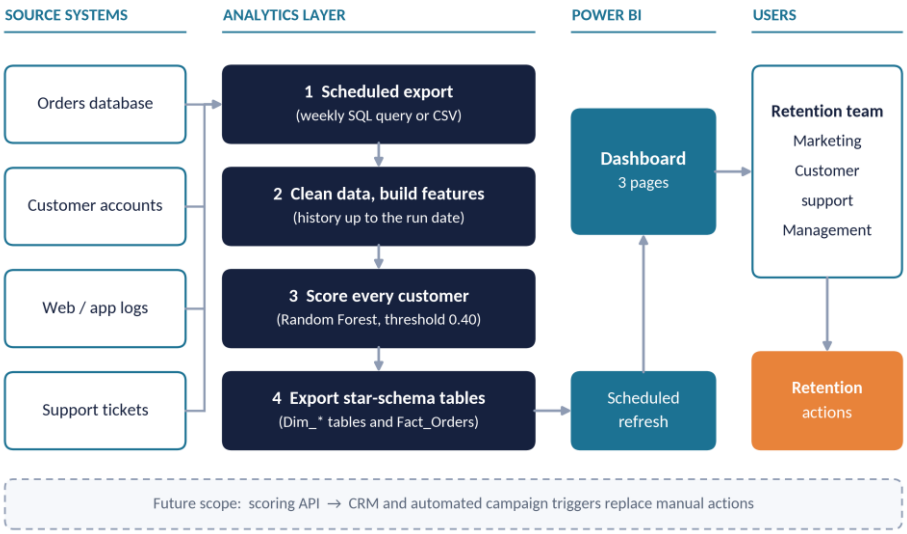


Figure 7.1: Proposed batch deployment on a live e-commerce store

Only two things change compared with the prototype. First, the synthetic files are replaced by scheduled exports of orders, customer accounts, web or app activity and support tickets from the store's database. Second, the existing scripts run automatically each week. Scoring needs only features up to the run date, so no churn label is required at that point; the model is retrained periodically, for example quarterly, when enough customers have passed through their churn window to provide fresh labelled outcomes.

The scored tables are published to Power BI Service with a scheduled refresh, and the retention team uses the dashboard to decide whom to contact. Real deployment requires real customer history and the store's permission to use it, which is why a synthetic dataset was used here, and it must comply with applicable data-protection requirements. The retention actions in this arrangement are manual; real-time scoring and automated triggers are future scope.

CHAPTER 8
CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This project built a complete churn-analytics pipeline for e-commerce: synthetic data generation, a leakage-safe snapshot labelling design, a comparison of three classifiers and a three-page Power BI dashboard on a star-schema model. Random Forest with a threshold of 0.40 was selected after the threshold analysis showed that it matches the recall of Logistic Regression with better precision, and it reached an AUC-ROC of 0.769 on held-out data. The dashboard reports a predicted churn rate of 35.26 per cent, 450 high-risk customers and $395.89K of revenue at risk. It shows that churn is concentrated in the At Risk and Lost segments, while the largest revenue exposure lies with Potential Loyalists and Loyal customers.

The combination of Python ML and Power BI has three practical benefits. It turns reactive reporting into forward-looking risk scoring; it delivers the output of the model in a governed, self-service form that business users can use without coding; and it expresses risk in financial terms, the language management uses to prioritise investment.

The project also showed the value of independent verification. Defects in the cutoff date, the handling of unscored customers and an evaluation table were caught only by recomputing results from raw evidence, and correcting them materially changed the figures. Within the stated limitations, chiefly the synthetic data and the cold-start population, the system shows how a data-science model can be embedded in everyday business decision-making.

8.2 Future Scope

- **Real data and out-of-time validation:** validate the pipeline on real transaction data, and evaluate the model on a later period than the one it was trained on, to obtain a realistic estimate of performance.
- **Scoring new customers:** build a separate model from signup information and early activity to give a risk estimate for customers who cannot be scored at present.
- **API integration and automated alerts:** wrap the model in a REST API for on-demand scoring from the order-management or CRM system, and connect high-risk segments to automated marketing triggers.
- **Lifetime-value forecasting:** replace the heuristic lifetime value with a probabilistic model such as BG/NBD with a Gamma-Gamma spend model [15] to forecast 12-month customer value.
- **Other enhancements:** probability calibration and segment-specific thresholds; sequence models on purchase histories; scheduled refresh and row-level security in Power BI Service; and an A/B test to measure the retention uplift of model-driven interventions.

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APPENDIX 1

Project Folder Structure and Execution Order

The project is organised as follows. The notebooks are run in numerical order; each uses the modules in the src folder and writes its output to the folders shown.

```
Minor Project/
  data/raw/             customers, orders, behaviour tables
  data/processed/       customer_features.csv
  data/star_schema/     Dim_Customer, Dim_Date, Dim_Product,
                        Dim_SupportTicket, Fact_Orders
  src/                  data_generation.py, cleaning.py,
                        features.py, churn_label.py,
                        model.py, star_schema.py
  notebooks/            01_eda.py
                        02_feature_engineering.py
                        03_model_training.py
                        04_explainability.py
                        05_star_schema_export.py
  models/               trained model, evaluation metrics,
                        feature importance
  reports/              EDA, evaluation and DAX summaries, plots
  requirements.txt
  README.md
```

Re-running the pipeline requires installing the packages in requirements.txt and running the five notebook scripts in order. All random processes use the fixed seed 42, so the results are reproducible.